LEARNER GUIDE

Order of operations and calculator judgement

Number Systems and Calculator Judgement · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The **order of operations** tells people and calculators what to do first: brackets, then powers and roots, then × and ÷ from left to right, then + and – from left to right. A **fraction bar** and a √ **sign** group things just like brackets do — but you cannot see those brackets. A calculator does exactly what it is *asked*. When you key a calculation in a line, you have to type the hidden brackets yourself.

What you are learning to notice

- A fraction bar groups the whole numerator and the whole denominator: key (6 + 9) ÷ (2 + 3), not 6 + 9 ÷ 2 + 3.
- A √ sign reaches over everything under its bar: key √(16 + 9), not √16 + 9.
- A power belongs only to what is just before it: $–4^2 = –16$, but $(–4)^2 = 16$.
- × and ÷ are worked from left to right, so 12 ÷ 2 × 3 = 18. To divide by 2 × 3, key 12 ÷ (2 × 3).
- Estimate before you trust the display. A square is never negative, and √ means the positive square root.

Moving through it, one step at a time

Before keying, look for the hidden groups: a fraction bar, a √ bar, a negative number being squared. Put each group in brackets. Press = and compare the answer with a quick estimate. If they disagree, look again at the keys.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — What the calculator heard

Work out the fraction with numerator 6 + 9 and denominator 2 + 3.

1. Keyed as it looks: 6 + 9 ÷ 2 + 3	the calculator works out 9 ÷ 2 first
2. It heard 6 + 4.5 + 3.	= 13.5
3. The fraction bar groups each part, so key brackets: (6 + 9) ÷ (2 + 3)	15 ÷ 5
4. Estimate check: 15 ÷ 5 is small, not 13.5.	= 3

The answer is 3. Each hidden group needed its own brackets.

Example 2 — A minus sign and a square

Compare -4^2 with $(-4)^2$.

1. -4^2: the square belongs to the 4 only. Square first.	$4^2 = 16$
2. Then apply the minus sign.	$-4^2 = -16$
3. $(-4)^2$: the brackets make -4 the number being squared.	-4×-4
4. A negative times a negative is positive.	$(-4)^2 = 16$

$-4^2 = -16$ but $(-4)^2 = 16$. To square a negative number, key it in brackets.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the order of operations, squares and square roots), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.